MIL-G-50973B (AR)
03 November 1983
SUPERSEDING
MIL-G-50973A (PA)
15 July 1975

MILITARY SPECIFICATION

GRENADES, GP: M42 AND M46 TAPE-STIFFENER ASSEMBLY FOR

This specification is approved for use by the U.S. Army Armament Munitions and Chemical Command, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements, provides quality assurance provisions, and packaging for one type of assembly designated as Tape-Stiffener Assembly for use with Grenades, GP, M42 and M46.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-5625	-	Webbing, Textile, Nylon, Tubular
MIL-A-48078	-	Ammunition Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-1169	-	Packaging, Packing and Marking for Shipment of Inert Ammunition Components

FSC 1330

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:46Z
Check the source to verify that this is the current version before use.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS (See 6.4)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

9215361 - Tape-Stiffener Assembly

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts. The parts shall comply with all requirements specified on Drawing (dwg. 9215361), all associated drawings, and with all requirements specified in applicable specifications.

3.3 First article testing. This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

3.4 Workmanship. All parts and assemblies shall be fabricated and finished in a thorough, workmanlike manner. They shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion products, and other foreign matter. In addition, plastic component parts shall be free from splits, cold flow, shrinkage, flash, gouges, cold shots, inclusion, porosity, or any similar characteristics. All marking and stamping shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:46Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection.

b. Quality Conformance Inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the contracting officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

PART DESCRIPTION	DRAWINGS	QUANTITY
Tape	9215379	50
Stiffener	9215261	(a)
Tape Stiffener Assembly	9215361	50

(a) Two (2) samples from each mold cavity.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:46Z
Check the source to verify that this is the current version before use.

TABLE I. First Article Inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-G-50973B (AR)

PARAGRAPH CATEGORY	TITLE / EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	DRAWING NUMBER / NEXT HIGHER ASSEMBLY / PARAGRAPH REFERENCE /INSPECTION METHOD
	Tape Stiffener Assembly		SHEET 1 OF 1		DRAWING NUMBER: SEE BELOW
	Tape (Dwg. 9215379)				
	Examination for defects	50	--	3.2	4.4.2.1
	Stiffener (Dwg. 9215261)				
	Examination for defects	(a)	--	3.2	4.4.2.2
	Tape Stiffener Assembly (Dwg. 9215361)				
	Examination for defects	50	--	3.2	4.4.2.3
	Parallelism under load test	50	--	3.2	4.5.1

NOTES:

(a) Two (2) samples from each mold cavity.

DRDAR-QA Form 160 Jul 77

Source: https://assist.dla.mil -- Downloaded: 2026-02-22T20:46Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:46Z
Check the source to verify that this is the current version before use.

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.1	Tape		9215379
			NEXT HIGHER ASSEMBLY 9215361

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101	Residual shrinkage test	a		3.2	4.5.3
102	Tape overlap fuzed improperly		0.40%	3.2	Visual
MINOR:					
201	Width		0.65%	3.2	Gage
202	Maximum overlap distance between ends of tape		0.65%	3.2	Gage
203	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

a. Three (3) samples per lot of tape

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:46Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-G-50973B (AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.2	Stiffener		9215261
			NEXT HIGHER ASSEMBLY 9215361

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined	See 4.5.2			
MAJOR:					
101	Symmetry of leg width with width of vertical cutout (4 places)		0.40%	3.2	Gage
102	Symmetry of narrowest section at top with width of vertical cutout		0.40%	3.2	Gage
103	Symmetry of slot length with width of vertical cutout		0.40%	3.2	Gage
104	Large inside radius connecting legs		0.40%	3.2	Gage
105	Distance across legs at bottom		0.40%	3.2	Gage
106	Flash excessive or in areas not allowed by the drawing		0.40%	3.2	Visual
107	Gating size or location improper		0.40%	3.2	Visual
108	Width improper		0.40%	3.2	Gage
MINOR:					
201	Total height		0.65%	3.2	Gage
202	Total width of legs		0.65%	3.2	Gage
203	Mold or cavity identification missing or incorrect		0.65%	3.2	Visual
204	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

Source: https://assist.dla.mil -- Downloaded: 2026-02-22T20:46Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-G-50973B (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9215361
4.4.2.3	Tape-Stiffener Assembly				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL:	None defined				
MAJOR:					
101	Parallelism under load		0.40%	3.2	4.5.1
102	Tear, twist or fold in tape		0.40%	3.2	Visual
103	Distance between fuzed area and stiffener when assembled in accordance with Note 3 of dwg. 9215361		0.40%	3.2	Gage
104	Tape not completely inserted in slots of stiffener		0.40%	3.2	Visual
105	Crack(s) in stiffener		0.40%	3.2	Visual
MINOR:					
201	Evidence of poor workmanship		0.65%	3.4	Visual

NOTES:

DRSMC-QA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:46Z
Check the source to verify that this is the current version before use.

4.4.3 Testing. This section is not applicable to this specification.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection.

4.5.1 Parallelism of tape-stiffener assembly under load. The test shall be performed using equipment in accordance with 4.4.4. The tape-stiffener assembly shall be mounted in a fixture. The load specified on the applicable drawing shall be applied to the assembly and the parallelism shall be measured. If any assembly fails to comply with the requirement specified on the applicable drawing, the assembly shall be classified defective and removed from the lot (non-destructive test).

4.5.2 Dimensional control of molded parts. In place of the normal sampling associated with the Classification of Defects, and with the approval of the Contracting Officer, and after a curing time for the parts has been approved (see Note), a sample of three (3) parts (as molded) from each cavity may be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of each cavity during production.

The inspection of the parts shall be performed only when the curing time for the parts has been reached. The inspection for control of each cavity during production shall be performed on the above quantities from each cavity after production of each 5000 parts or at the end of each shift, providing the number of parts produced during that shift is less than 5000 parts. Of the three (3) samples, one (1) sample shall be the last part produced. Parts that fail to comply with dimensional requirements after dimensional stability has been attained shall be classed defective.

If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for corrective action.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:46Z
Check the source to verify that this is the current version before use.

All cavities formerly removed from production because of some fault may, after reworking, be returned to production providing they pass the qualification test as above. The contractor may request a change of inspection frequency providing he presents objective evidence to substantiate the request to the Contracting Officer.

NOTE: In establishing a curing time dimensionally check ten (10) parts from each cavity at periodic intervals (e.g., every 30 min., 60 min., etc.), until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to Commander, US Army Armament Research and Development Center (ARDC), ATTN: DRSMC-QAR-Q, Dover, NJ 07801. If there is a change in material, cycle time, or if a cavity is reworked, or a new cavity is used, a new curing time shall be established and approved.

4.5.3 Residual shrinkage test. The test shall be performed using equipment in accordance with 4.4.4. The tape shall be tested in accordance with MIL-W-5625, paragraph 4.3.3. If any tape fails to comply with the requirement specified on the applicable drawing, the lot shall be rejected.

5. PACKAGING

5.1 Preservation.

5.1.1 Level B. The Tape-Stiffener Assembly (dwg. 9215361) shall be preserved and packaged in accordance with MIL-STD-1169. The bulk packaging arrangement may be utilized.

5.2 Packing.

5.2.1 Level B. The Tape-Stiffener Assembly (dwg. 9215361), preserved and packaged as specified in 5.1, shall be packed for shipment in accordance with MIL-STD-1169.

5.3 Marking. Marking of the carton shall comply with MIL-STD-1169.

5.4 Shipping. When assemblies from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. This item is intended for use with the Grenades, GP; M42 and M46.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:46Z
Check the source to verify that this is the current version before use.

6.2 Ordering data. See MIL-A-48078.

6.3 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs as required to Commander, US Army Armament Research and Development Center (ARDC), ATTN: DRSMC-QAR-I, Dover, NJ 07801.

6.4 Drawings. Drawings listed in Section 2 of this specification under the headings U.S. Army Armament Research and Development Center (ARDC) may also include drawings prepared by and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal, or ARRADCOM drawings. Technical data originally prepared by these activities is now under the cognizance of AMCCOM.

6.5 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1330-A195)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:46Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-50973B (AR)	Grenades, GP: M42 and M46 Tape-Stiffener Assembly For

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:46Z
Check the source to verify that this is the current version before use.